Hamiltonian transformation for accurate and efficient band structure interpolation
Supplemental Material

Kai Wu,[1,2] Yingzhou Li,[3] Wentiao Wu,[1] Lin Lin,[4,5] Wei Hu,[1,*] and Jinlong Yang[1,†]

[1] *School of Future Technology, University of Science and Technology of China, Hefei, Anhui 230026, China*
[2] *Dipartimento di Fisica, Università di Roma Tor Vergata, Via della Ricerca Scientifica 1, 00133 Rome, Italy*
[3] *School of Mathematical Sciences, Fudan University, Shanghai 200433, China*
[4] *Department of Mathematics, University of California, Berkeley, California 94720, United States*
[5] *Applied Mathematics and Computational Research Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, United States*

S1. RANDOMIZED QRCP

Let N_r, N_b, and N_k denote the numbers of real-space grids, bands, and $\mathbf{k}$-points, respectively. The Bloch wavefunctions at different $\mathbf{k}$-points are concatenated into a large matrix Ψ of size $N_r \times N_b N_k$. However, when N_k is large, performing low-rank decomposition algorithms directly on Ψ becomes inefficient. To address this issue, the time-consuming QR with column pivoting (QRCP) and QR decompositions are applied to two smaller matrices, Ψ_1 and Ψ_2, instead of the large matrix Ψ. Algorithm 1 outlines the decomposition process:

$$\Psi_{i\mathbf{k}}(\mathbf{r}) = \sum_{\mu=1}^{N_\mu} Q_\mu(\mathbf{r}) C_{i\mathbf{k}}^{\mu}. \tag{S1}$$

To construct the smaller matrix Ψ_1, we estimate the maximum rank of Ψ as r_e, randomly select $1.5r_e$ columns from the large matrix Ψ and record their indices as *cols*. We then left-multiply this subset of Ψ by a random matrix L of size $r_e \times N_r$, whose elements are drawn from a unit-variance Gaussian independent and identically distributed (GIID) distribution:

$$\Psi_1 = L\Psi(:, \mathit{cols}). \tag{S2}$$

The resulting matrix Ψ_1 has dimensions $r_e \times 1.5r_e$. We perform a QR decomposition with column pivoting (QRCP) on Ψ_1:

$$\Psi_1 \Pi = Q_1 R_1. \tag{S3}$$

Here, Q_1 is an $r_e \times r_e$ matrix with orthonormal columns, R_1 is an upper triangular matrix, and Π is a permutation matrix that reorders the columns of Ψ_1. In practice, setting $r_e = 40N_b$ is sufficient for small systems, while r_e can be reduced to $20N_b$ for larger systems. Note that r_e is independent of N_k.

The permutation Π arranges the columns so that the magnitudes of the diagonal elements of R_1 are in non-increasing order. These diagonal elements indicate the importance of the corresponding columns in Q_1. We define N_μ as the number of diagonal elements in R_1 that are larger than a threshold σ. This number represents the numerical rank of the space spanned by the lowest N_b orbitals across all $\mathbf{k}$-points. Importantly, N_μ tends to remain a constant multiple of N_b, even as N_k approaches infinity.

After determining the indices of the most representative columns from Ψ_1, we construct Ψ_2 by extracting the corresponding columns from the large matrix Ψ:

$$\Psi_2 = \Psi(:, \mathit{cols})\Pi(:, 1:N_\mu), \tag{S4}$$

where $\Pi(:, 1:N_\mu)$ selects the first N_μ columns of the permutation matrix Π. We then perform a QR decomposition on Ψ_2:

$$\Psi_2 = QR_2. \tag{S5}$$

* whuustc@ustc.edu.cn
† jlyang@ustc.edu.cn

Ψ is rank-redundant, and Ψ_2 captures the most representative columns of Ψ, thus both matrices share the same basis set Q. We can recover the wavefunction coefficients C in the basis of Q using:

$$C = Q^\dagger S \Psi, \tag{S6}$$

where S is the overlap matrix of Ψ. This allows us to express the original wavefunction Ψ in terms of the orthonormal basis provided by Q.

Algorithm 1: Randomized QRCP for Hamiltonian Transformation

Input : Eigenvectors $\{\psi_{i\mathbf{k}}(\mathbf{r})\}$, estimated max rank r_e, rank threshold σ
Output: Numerical basis set Q, wavefunction C
1. Concatenate $\{\psi_{i\mathbf{k}}(\mathbf{r})\}$ into a $N_r \times N_b N_k$ matrix: $\Psi = [\psi_{\mathbf{k}_1}(\mathbf{r}), \ldots, \psi_{\mathbf{k}_{N_k}}(\mathbf{r})]$;
2. Randomly select $1.5r_e$ columns from Ψ, record indices as $cols$;
3. Construct a $r_e \times N_r$ random matrix L;
4. Compute $\Psi_1 = L\Psi(:, cols)$;
5. **Perform QRCP** on Ψ_1: $\Psi_1 \Pi = Q_1 R_1$;
6. Take N_μ as the number of diagonal elements of R_1 larger than σ;
7. Compute $\Psi_2 = \Psi(:, cols)\Pi(:, 1 : N_\mu)$;
8. **QR decompose** Ψ_2: $\Psi_2 = QR_2$;
9. Calculate $C = Q^\dagger S \Psi$;

S2. THE CHOICE OF HT PARAMETERS

Here, ε represents the maximum eigenvalue in the SCF calculation, and $\mathrm{erf}(x)$ is the error function. The function f has two adjustable parameters, a and n. The parameter $a \geq 0$ controls the width of the transition region (typically set in proportion to the energy range of the entangled bands), while n governs the smoothness of the function f; a larger n results in a smoother function. The formula of f is obtained by integral from f' with $f(\varepsilon) = 0$, which is shown in Eq. (**??**).

S3. THE SIZE OF N_μ

We conduct tests to explore the relationship between the size of numerical basis set N_μ, the size of real space grids N_r, the number of bands N_b, and the number of Wannier functions N_w. The system used is silicon, with the generalized gradient approximation of the Perdew-Burke-Ernzerhof (GGA-PBE) [1] functional. The cutoff energy is set to 100 Ry, and $n = 3$ is used for the transform function. The SCF FFT grid is $48 \times 48 \times 48$. In HT, we adopt a coarser grid for orbitals, set to $25 \times 25 \times 25$, yielding $N_r = 15625$. Although we do not perform WI, the mean absolute error (MAE) calculation considers the lowest 8 bands, effectively setting $N_w = 8$.

In the QRCP process, we sort the diagonal values of R by their absolute magnitudes, with the largest normalized to 1, and ignore values smaller than ϵ_t. The relationship between ϵ_t, N_μ, N_k, and MAE is shown in Fig. S1. When ϵ_t is fixed, N_μ converges to a constant as N_k becomes sufficiently large. Specifically, for $N_k = 27$, we find $N_\mu \approx 90$ achieves maximum accuracy, and for $N_k = 1000$, $N_\mu \approx 225$ reaches maximum accuracy. In these cases, we observe $N_\mu \approx 0.01N_r$ and $N_\mu \approx 20N_w$. In the QRCP procedure, we include 14 bands, i.e., $N_b = 14$. If N_b is reduced to 10, the required N_μ decreases. The ratio N_μ/N_r also depends on the cutoff energy, with higher cutoff energies corresponding to smaller values of N_μ/N_r. Based on our experience, $N_\mu/N_r \approx 0.01 \sim 0.1$ in most cases.

[1] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, Phys. Rev. Lett. **77**, 3865 (1996).

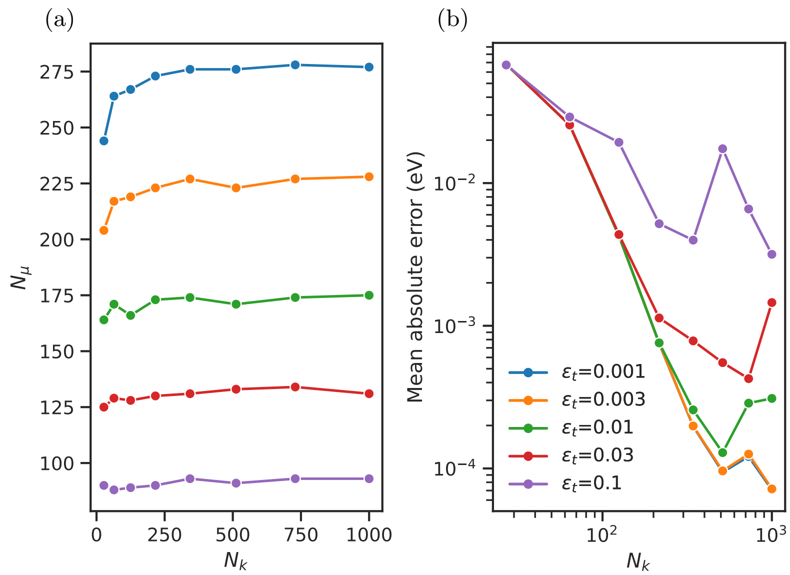

Fig. S1. The relationship between ϵ_t, N_μ, N_k and MAE.